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Biochemistry Module, 1st Year Medicine

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General characteristics of lipids

Course objectives:

-List the solubility characteristics common to all lipids.

-Describe the main biological roles played by lipids.

-Classify fatty acids according to their number of carbon atoms and double bonds.

- Characterize the spectral, solubility, melting, boiling and emulsification properties of fatty acids.

- Specify the two chemical properties of fatty acids related to their carboxyl group.

- Define the chemical properties of fatty acids related to the presence of double bonds.

Academic Year 2025-2026

I/Definition

- This is a group of molecules that are chemically heterogeneous.
- They are defined on the basis of a common physical characteristic:

They are poorly soluble in water, but soluble in organic solvents (ether, benzene, chloroform, etc.).

- Lipids are:

- Either hydrophobic, if they contain only non-polar groups.
- Or amphiphilic (or amphipathic), if they contain both polar and non-polar groups with a polar head (hydrophilic) and a hydrophobic tail (carbon chain).

- The terms oil, butter, fat and wax only refer to their physical state, liquid or solid, at ambient temperature.

II/ Classification of lipids:

A/ True lipids:

Result from the condensation of fatty acids with alcohols.

1- Simple lipids

Composed of C, H, and O, and include:

- **Glycerides:** the alcohol is glycerol
- **Cerides:** the alcohols are long-chain (waxy)
- **Sterides:** the alcohol is a sterol (polycyclic)

2- Complex lipids:

Composed of C, H, O as well as N, S, P, etc.

These include **glycerophospholipids, sphingolipids**, and **sulfated sterols**.

More complex structures exist, such as **glycolipids** (which contain sugars).

B/ Lipid-like compounds (lipoids):

✓ **Isoprenoids:** derived from isoprene units (5 C); this category includes sterol derivatives and the fat-soluble vitamins A, D, E, and K.

✓ **Eicosanoids:** mediators derived from fatty acids.

Example: Prostaglandins (these are cytokines), etc.

III/ Biological roles of lipids

In the body, lipids serve several important biological functions:

– Structural role, such as phospholipids in cell membranes,

– Energy storage role, such as fats,

– Mediator role, such as hormones,

– Role in metabolism (cofactors, vitamins, etc.).

Knowledge of lipid molecules is essential in medicine. Significant examples include atherosclerosis—a disease characterized by cholesterol deposition in the intima of arteries—as well as issues related to nutrition and dietetics.

FATTY ACID STRUCTURES AND PROPERTIES

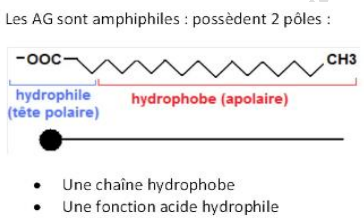
I/Definition:

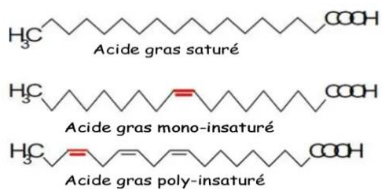
Fatty acids (FA) are carboxylic acids R-COOH in which the radical R is an aliphatic hydrocarbon chain of variable length that gives the molecule its hydrophobic character.

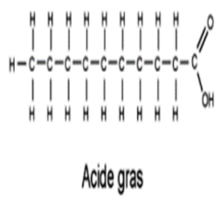
The vast majority of natural fatty acids share the following characteristics:

- linear chain with an even number of carbon atoms

- saturated or partially unsaturated with a maximum of 6 double bonds.







II/Classification:

The classification of fatty acids is based on:

- ❖ The number of carbon atoms:

n is between (4 and 36), numbered from the carboxyl carbon atom.

n is almost always even; the smallest FA is **C4**: **butyric acid** (it produces the smell of rancid butter and body odor...).

Most **natural FA** have between **14 and 24** carbon atoms.

❖ The number of double bonds, we distinguish between:

•**Saturated fatty acids** (without double bonds), whose molecules are both:

-flexible (total freedom of rotation around each **C–C** bond), and

-stretched (most stable conformation).

General formula for saturated fatty acids : CH3-(CH2)(n-2)-COOH

(CnH2nO2).

Example of a saturated fatty acid:

C16: palmitic acid (n-hexadecanoic acid)

C16H32O2 with the formula CH3-(CH2)14–COOH.

•**Unsaturated fatty acids** (with one or more double bonds) are said to be

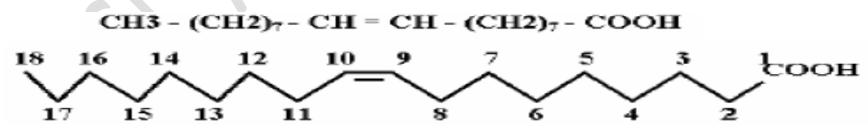
mono- or polyunsaturated, as the double bond creates a rigid 30° bend in the molecule.

For polyunsaturated fatty acids, the double bonds are generally nonconjugated, i.e. they are separated by a **CH2** group.

…….-CH2-CH=CH-CH2-CH=CH-CH2-..........

-Example of a monounsaturated fatty acid:

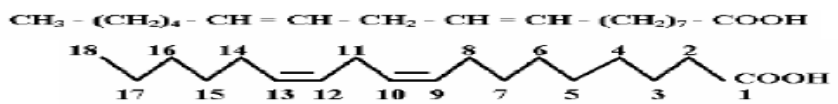
C18 oleic acid has a double bond in position 9. It is a (non-essential) fatty acid that is very abundant in vegetable and animal fats.



-Examples of polyunsaturated fatty acids:

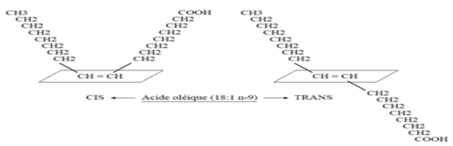
C18 linoleic acid has two double bonds at positions 9 and 12

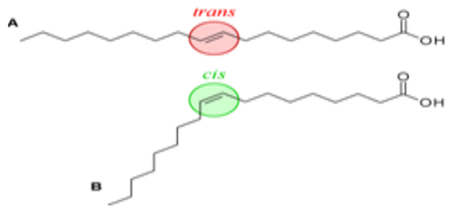
and is an omega-6 fatty acid.



The presence of a double bond gives the aliphatic chain two possible configurations: **the cis (Z)** configuration and the **trans (E)** configuration.

If they are on the **same side**, the bond is called **cis;** if they are **above** and **below**, the bond is called **trans**. Most **natural fatty acids have a cis (Z)** configuration.





III/Nomenclature

There are three types of fatty acid nomenclature:

a) International nomenclature:

This is the chemical nomenclature of the molecule, characterized by:

-The addition of the **anoic** radical for saturated fatty acids;

-The addition of the **enoic** radical, the positions of the double bonds, and their spatial configuration (cis) and (trans) for unsaturated fatty acids.

Numbering starting from the carboxyl group COOH (always denoted 1), with the other carbons bearing their sequential numbers.

The symbol used:

for saturated fatty acids is **Cn:0** and

for unsaturated fatty acids **Cn: mΔ (p, p'….).**

- n= number of C atoms
- 0 = absence of double bonds
- m= number of double bonds
- p, p' …= positions of double bonds

b) Usual nomenclature

Each fatty acid is given a specific name, generally based on its discovery.

Example: -the 16C saturated fatty acid is called palmitic acid, from the Latin palmus **(palm),**

-the 12C saturated fatty acid is called lauric acid **(laurel)**

c) Physiological nomenclature (omega)

-Used mainly by nutritionists.

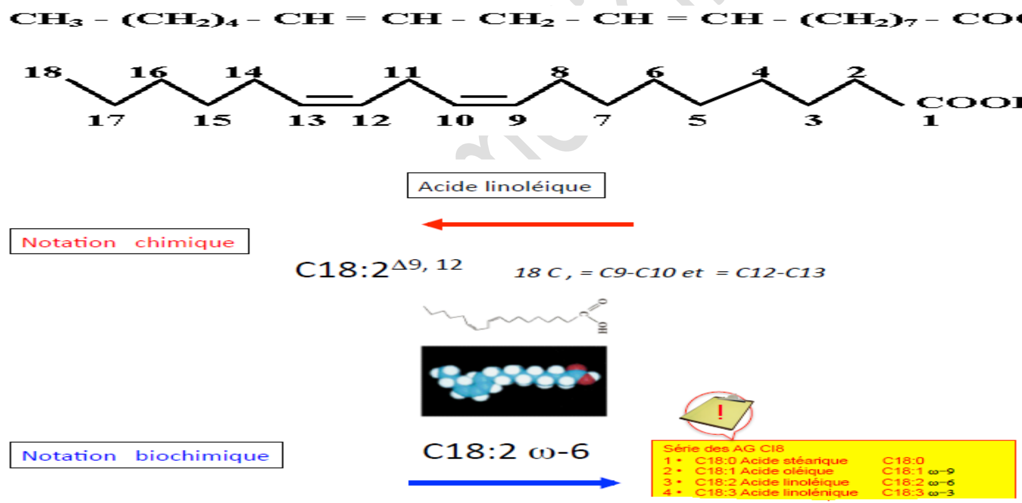
-Only applies to unsaturated fatty acids.

-It takes into account the first double bond encountered, but starts counting from the methyl group (CH3).

-It allows fatty acids to be identified by family.

Symbol **Cn: mωp** where:

- n= number of C atoms
- m = number of double bonds
- p= position of the first double bond



Some important fatty acids:

-Palmitic acid C16:0

-Palmitoleic acid C16:1Δ^9

-Stearic acid C18:0

-Oleic acid C18:1Δ^9

-Linoleic acid C18:2Δ^9,12 — they are essential for humans,

-Linolenic acid C18:3Δ^9,12,15 — they must be obtained from food,

they are called omega-6 and omega-3

Atypical fatty acids: FAs can be branched, cyclic, or carry functions other than the acid function.

ACIDES GRAS	NOMBRE D'ATOMES DE CARBONE	FORMULE CHIMIQUE	SOURCE
Saturés			
Acide butyrique	4	C_3H_7COOH	Beurre
Acide caproïque	6	$C_5H_{11}COOH$	Beurre
Acide caprylique	8	$C_7H_{15}COOH$	Noix de coco
Acide caprique	10	$C_9H_{19}COOH$	Huile de palme
Acide laurique	12	$C_{11}H_{23}COOH$	Noix de coco
Acide myristique	14	$C_{13}H_{27}COOH$	Huile de muscade
Acide palmitique	16	$C_{15}H_{31}COOH$	Graisses
Acide stéarique	18	$C_{17}H_{35}COOH$	Graisses
Acide arachidique	20	$C_{19}H_{39}COOH$	Huile d'arachide
Monoinsaturés			
Acide palmitoléique	16	$C_{15}H_{29}COOH$	Beurre
Acide oléique	18	$C_{17}H_{33}COOH$	Huile d'olive
Polyinsaturés			
Acide linoléique	18	$C_{17}H_{31}COOH$	Huile de lin
Acide linolénique	18	$C_{17}H_{29}COOH$	Huile de lin
Acide arachidonique	20	$C_{19}H_{31}COOH$	Huile d'arachide

The main fatty acids

IV/ Physicochemical properties of FA

A/ Physical properties

These are mainly determined by the length and degree of unsaturation of the carbon chain.

❖ **Melting point:**

This is the temperature at which the FA exists in liquid form.

It varies according to two parameters: the **number** of **C** atoms and the degree of **unsaturation**.

* The greater the number of C atoms, the higher the melting temperature.

* An increase in the number of double bonds leads to a decrease in the melting temperature.

-They are liquid if **n < 10 C**

Solid if **n > 10 C**

❖ **Density:**

The density of FA is low; oil floats on water.

❖ **Spectral properties**

-Saturated FA do not absorb light in the UV range; unsaturated FA do not absorb light in either the visible or UV range at 280 nm because the double bonds are not conjugated.

-Polyunsaturated FA heated in an alkaline medium isomerize into FA with conjugated double bonds and absorb in the UV range.

❖ **Solubility:**

FAs have a hydrophilic pole (-COOH) and a hydrophobic pole (-R).

- Only short-chain FA (C4, C6) are soluble in water.

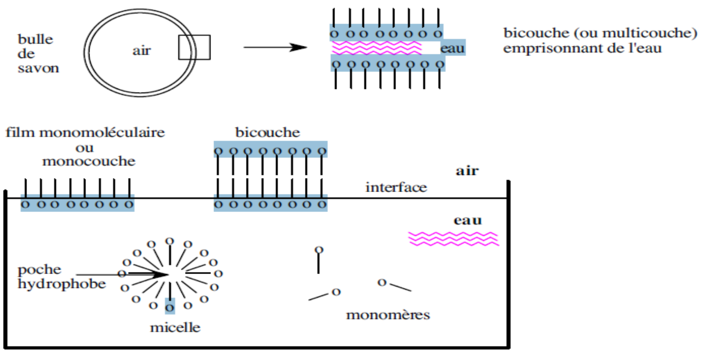
The others have too long a radical and their nonpolar character outweighs their polar character.

- Double bonds reduce the nonpolar character.

If the fatty acids are on the surface, they form a molecular film (single or double layer, or multi-layer).

If they are stirred vigorously in water, micelles are created = creation of an emulsion.

To solubilize most fatty acids, nonpolar organic solvents (ether, benzene) are used.



B/Chemical properties

These depend on the presence of the -COOH group, the possible presence of double bonds, and the possible presence of other radicals. The hydrocarbon chain has no particular chemical properties.

1/Properties due to the –COOH function

- **Formation of alkaline salts = Soaps**

In the presence of a base (KOH, NaOH), FAs produce salts commonly known as SOAPS.

$$R\text{-}COOH + NaOH \longrightarrow R\text{-}COO\text{-}Na^{+} + H_2O$$

- Sodium soaps are hard
- Potassium soaps are soft

Industrially, soaps are prepared by saponification of glycerides.

-Acid or saponification index (SI): by definition

SI = mass of potash, in **mg**, required to neutralize the free acidity contained in

1g of fat.

Properties of soaps:

-Wetting property:

-Foaming property: Soaps can trap air within micelles.

-Emulsifying property: Soaps coat hydrophobic substances (such as oil) inside stable micelles.

- **Formation of heavy metal salts**

Soaps can be precipitated in the presence of heavy metal salts (Ca^{2+}, Mg^{2+}, etc.)

$$2R\text{-}COO^{-}, Na^{+} + Ca^{2+} \longrightarrow (RCOO)_2Ca + 2Na^{+}$$

Application: when water is rich in Ca^{2+} (“hard” water), it is difficult to obtain foam in the presence of detergent.

- **Esterification:**

$$R\text{-}COOH + R'\text{-}OH\text{-} \longrightarrow R\text{-}(C{=}O)\text{-}O\text{-}R' + H_2O$$

AG + alcohol → ester

-This explains the formation of more **complex lipids**.

-There are **enzymes** that carry out this reaction.

2/Properties due to the presence of the double :

a) Addition reaction:

-Hydrogenation:

... CH2-CH=CH-CH2 ... + H2 $\longrightarrow$ ... -CH2-CH2-CH2-CH2-...

Application: process used to make margarine from oil. Margarine is more resistant to oxidation than oils.

-Halogenation:

Determination of the iodine value (IV). The number of double bonds in an FA is determined.

By definition, the iodine value is:

Number of **grams** of iodine that can be fixed by **100g** of fat

R-CH**=**CH-R'-COOH + **I2** $\longrightarrow$ R-CH**I**-CH**I**-R'-COOH

b) Oxidation reaction:

Oxidation by KMnO4 in an alkaline medium causes the fatty acid to **break** at the **double bond**, producing **two carboxylic acids**.

R-CH=CH-R'-COOH + KMnO4 $\longrightarrow$ R-COOH **+** HOOC-R'-COOH

An **acid** and a **diacid** are formed for each double bond.

This reaction, followed by analysis of the products formed, **allows** the position of the double bond in the molecule to be determined.

Oxidation by oxygen in the air causes fats to go rancid.

Intracellular enzymatic oxidation of **arachidonic acid (C20:0)** by cyclooxygenase (cyclization + oxidation) leads to **prostaglandins**, which are highly active mediators that degrade very quickly.

• **Biological action of prostaglandins**. They are involved in:

— the contraction of smooth muscles (intestines, uterus, blood vessels);

— the regulation of metabolism;

— platelet aggregation. The inhibition of platelet cyclooxygenase by aspirin is useful in therapy (antiplatelet agent).

c) Isomerization

Heating isomerizes the cis forms into thermodynamically more stable trans forms.